Complete Project Flow

From first visit to a validated, visual travel itinerary

THE PRODUCT

A ChatGPT-style Travel AI where users explore trips, compare travel options, receive interactive charts, and refine itineraries in conversation.

01 USER

Guest or signed-in visitor

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02 CHAT

Temporary or saved conversation

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03 FASTAPI

Validation, security, orchestration

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04 LANGGRAPH

Intent > Research > Plan > Validate

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05 TRAVEL DATA

Maps, providers, RAG knowledge

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06 RESULTS

Itinerary, charts, maps, booking links

User Journey & Authentication

Two chat modes, email OTP password recovery, simple navigation

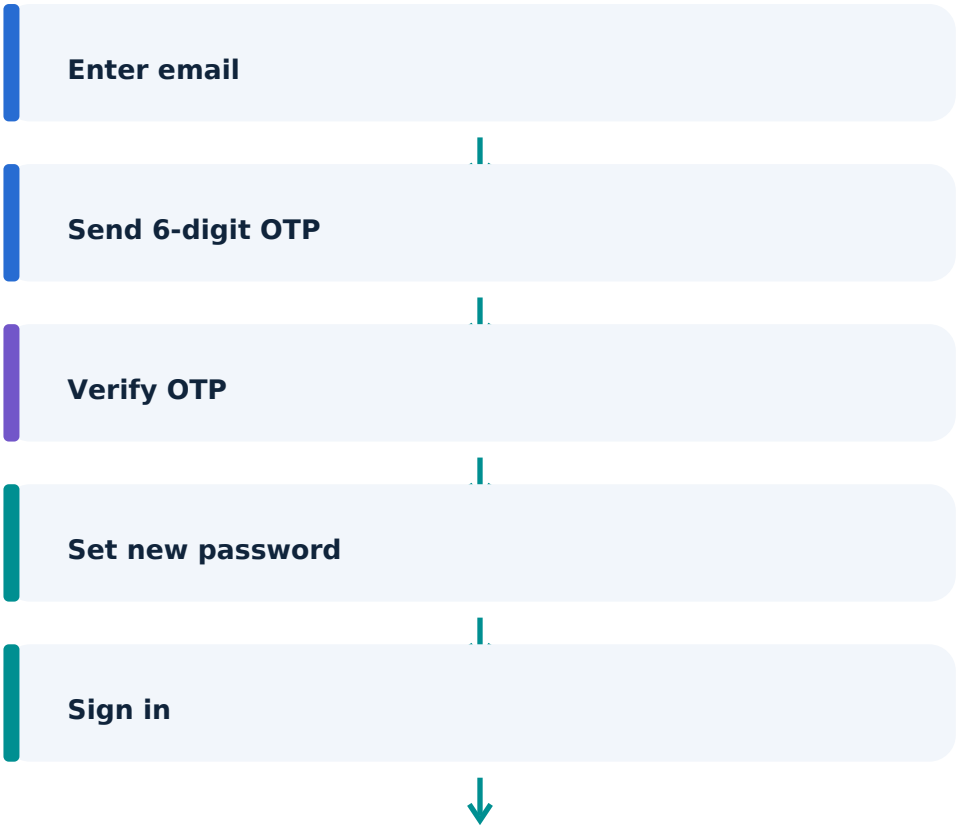
GUEST MODE

Open the website, explore directly, use temporary chat only. No account required.

SIGNED-IN MODE

Sign up / sign in; saved chat history in sidebar plus temporary mode.

PASSWORD RECOVERY - EMAIL OTP, NOT RESET LINK



SECURITY RULES

- OTP expires after 5 minutes; limited attempts and resend cooldown.
- Store password hashes and OTP verifiers; never plaintext credentials.
- Use a single-use, short-lived reset authorization after OTP verification.
- Secure session cookies and rate limits on public endpoints.

Agentic Trip Planning Workflow

Five LangGraph nodes with provider tools and deterministic checks

1 INTENT NODE

Extract origin, destinations, dates, budget, passengers, return deadline.

2 RESEARCH NODE

Query available maps, transport, hotel, attraction tools and RAG.

3 PLANNING NODE

Compare candidate routes, travel durations, stays and total costs.

4 VALIDATION NODE

Check schedule feasibility, office deadline, budget and source freshness.

5 RESPONSE NODE

Return typed JSON for itinerary, cards, maps, charts, and booking links.

CRITICAL RULE

LLM explains and orchestrates. Python rules + provider data validate dates, routes, prices and availability. Never invent live fares or seat availability.

Chat Persistence & Automatic Cleanup

Only two chat modes; no share-chat functionality

TEMPORARY CHAT

Available only in the active browser tab. No history entry. Short-lived backend state with automatic TTL cleanup.

SAVED CHAT

Requires sign-in. Visible in sidebar, can be reopened and continued. Expires 7 days after creation.

SAVED CHAT LIFECYCLE

01 User starts saved conversation

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02 PostgreSQL stores chat + itinerary

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03 Sidebar shows chat until expiry

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04 After 7 days: access immediately denied

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05 Cleanup worker deletes associated records

REAL-WORLD DELETION CONSIDERATIONS

Remove LangGraph checkpoints, queued state and related artifacts too. Align logs and backup retention with your deletion policy. Browser closing cannot guarantee instant server deletion.

Frontend Results & Data Contracts

The backend returns structured data; React renders engaging visuals

AI EXPLANATION

Natural-language suggestions and important warnings.

DAY-WISE ITINERARY

Timeline of destinations, activities and travel segments.

ROUTE MAP

Geographical stops and available routing information.

BUDGET DONUT CHART

Transport / hotel / food / activities / buffer.

TRANSPORT BAR CHART

Compare fares and travel durations across available modes.

HOTELS & ATTRACTIONS

Rich cards with timings, fees, provider info and booking links.

DATA QUALITY CONTRACT

Every quote should identify currency, provider, freshness and whether it is LIVE, ESTIMATED or UNKNOWN. React/Recharts visualizes the JSON; it must not fabricate prices. Booking redirects are not confirmed bookings.

Technical Architecture & Build Order

React + FastAPI + LangGraph + RAG + PostgreSQL + Redis + AWS

REACT + TAILWIND

Chat UI, authentication, visual results

FASTAPI

REST APIs, sessions, OTP, rate limits, SSE

LANGGRAPH

Research, planning, validation, follow-up

TRAVEL INTEGRATIONS

Maps, transport, hotels, attractions, RAG

POSTGRESQL + REDIS

Saved 7-day chats, temporary state, queues

DOCKER + AWS

Hosting, secrets, logs, cleanup, monitoring

IMPLEMENTATION ORDER

1. Foundation > 2. Auth + OTP > 3. Chat modes > 4. LangGraph > 5. Travel APIs > 6. Charts + UI integration > 7. Tests + deployment